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CareSync System Architecture Breakdown

1. **Frontend Layer (React.js)**

- **Web Dashboard**: This is the primary interface for doctors and admins to manage clinical resource tracking and medical inventory.
- **Patient Portal**: A mobile-responsive web app where patients book appointments and get their QR codes.
- **Live Display**: A specialized view using **WebSockets** for the clinic's front desk and patient-facing displays to sync the queue instantly.

2. **Backend Layer (Node.js & Supabase)**

- **API Logic (Node.js)**: Handles the "heavy lifting" like the **AI-Driven Triage** algorithms and generating PDF/CSV institutional reports for auditing.
- **Real-time Engine**: Utilizing **Supabase Realtime** (via WebSockets) to ensure that when a physician signs off on a result, the "Release Result" status updates immediately.
- **Security Layer**: Implementing **End-to-End Encryption** for patient records (EHR) to ensure data privacy.

3. **Data & Cloud Layer (Supabase/PostgreSQL)**

- **Relational Database**: Stores everything from vital signs to equipment maintenance logs.
- **Auth Service**: Manages secure logins for different staff roles.
- **Storage**: A secure bucket for storing digital courseware for students or medical documents for patients.